

MANGANESE INTELLIGENCE

AI-Powered Manganese Reserve Potential and Production Shortfall Intelligence System

Smart India Hackathon 2026

Problem Statement: 26009

Team: Mine Matrix

Team Members

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1. Executive Summary

Manganese Intelligence is an AI/ML-powered mining decision-support prototype developed for Smart India Hackathon 2026 Problem Statement 26009.

The system is designed to combine multiple mining-related data sources, including geological information, production records, equipment performance, equipment downtime, blasting delays, weather information and satellite-derived indicators.

The platform provides a centralized web-based dashboard through which users can monitor production performance, identify potential production shortfalls, analyze operational risks, monitor equipment risks, visualize reserve-potential zones and evaluate operational what-if scenarios.

The system follows the workflow:

Data → Analysis → Prediction → Risk Assessment → Root Cause → Recommendation → Decision Support

The prototype is designed to demonstrate how AI/ML and space-derived information can support data-driven mining operations.

The current prototype uses synthetic/sample data for demonstration. It should therefore be treated as a decision-support prototype and not as a validated production system for actual mining operations.

2. Problem Statement

Problem Statement ID

26009

Problem Statement

Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls.

3. Background

Mining operations generate information from several different sources.

Examples include:

- Geological surveys
- Production records
- Equipment records
- Equipment downtime
- Blasting operations
- Weather conditions
- Satellite observations
- Mine location data

When these datasets are analyzed separately, it becomes difficult to obtain a complete view of mining operations.

Production shortfalls may be influenced by several factors at the same time.

For example:

Equipment Downtime + Blasting Delay + Weather Conditions → Production Impact

Similarly, identifying potential manganese-bearing zones can require analysis of geological and surface/sub-surface indicators together with spatial information.

Manganese Intelligence is designed to bring these different sources together into a unified decision-support platform.

4. Objectives

The main objectives of the system are:

1. Integrate different mining-related datasets.
 2. Analyze historical production performance.
 3. Estimate production performance.
 4. Identify potential production shortfalls.
 5. Calculate operational risk.
 6. Identify important contributing factors.
 7. Monitor equipment-related risks.
 8. Visualize reserve-potential zones.
 9. Integrate geological and satellite-derived indicators.
 10. Provide AI/ML-assisted decision-support recommendations.
 11. Provide what-if operational simulation.
 12. Present important information through a centralized dashboard.
-

5. Proposed Solution

Manganese Intelligence provides a unified mining intelligence platform.

The system collects and processes mining-related information and converts it into useful analytical outputs.

The major components are:

5.1 Data Ingestion

The system supports importing production information through:

- CSV
- Excel
- PDF
- Manual data entry

The uploaded information is processed by the backend and stored in the database.

5.2 Production Intelligence

Production records contain information such as:

- Production date
- Planned production
- Actual production
- Downtime hours
- Blasting delay hours

The system uses these records to analyze production performance.

5.3 Production Shortfall Analysis

Production shortfall can be represented as:

Shortfall = Planned Production – Actual/Estimated Production

If the estimated production is lower than the planned production, the system identifies a potential shortfall.

5.4 Risk Analysis

The system analyzes operational factors and produces an operational risk assessment.

Risk levels can include:

- Low
- Medium
- High

The dashboard can display:

- Risk score
 - Risk level
 - Primary contributing cause
-

5.5 Equipment Risk

The equipment module provides information about equipment performance and operational risk.

It can help identify equipment that may require:

- Maintenance
 - Monitoring
 - Operational attention
 - Redeployment
-

5.6 Reserve Potential Analysis

The reserve module is designed to identify areas with relatively higher manganese reserve potential.

Potential inputs include:

- Geological samples
- Location
- Satellite-derived indicators
- Soil moisture
- Vegetation index
- Land surface temperature
- Other spatial indicators

The results can be displayed on an interactive map.

Important:

The system identifies **reserve potential**, not confirmed mineral reserves.

Actual reserve estimation requires validated geological exploration and domain-expert verification.

5.7 AI/ML-Assisted Recommendations

The system uses available operational information to generate decision-support recommendations.

Possible recommendation categories include:

- Maintenance prioritization
- Equipment redeployment
- Production scheduling adjustment
- Blasting schedule optimization
- Downtime reduction
- Risk mitigation

These recommendations are intended to assist human decision-makers rather than replace them.

5.8 What-if Simulation

The system provides an interactive simulation mechanism.

For example, a user can change:

Downtime Reduction = 30%

The system estimates the possible change in:

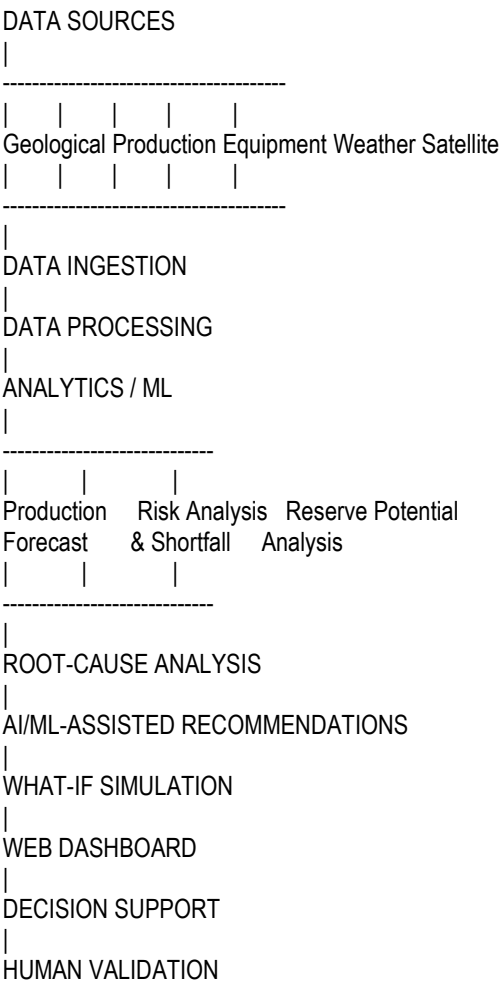
- Downtime
- Production
- Production shortfall

This allows users to explore hypothetical operational improvements.

The simulation is an analytical prototype and should not be interpreted as a guaranteed production outcome.

6. System Workflow

The complete system workflow is:



7. System Architecture

The application follows a layered architecture.

7.1 Frontend Layer

The frontend provides the user interface.

Current implementation uses:

- React
- Vite
- JavaScript
- Axios
- Charting components
- Interactive map functionality

The frontend communicates with the backend through REST APIs.

7.2 Backend Layer

The backend is implemented using:

- Python
- FastAPI
- Pydantic
- Uvicorn
- PostgreSQL connectivity

The backend handles:

- API requests
 - Data processing
 - Database communication
 - Production analysis
 - Equipment analysis
 - Reserve analysis
 - Recommendations
 - Simulation
 - Data ingestion
-

7.3 Database Layer

The application uses PostgreSQL.

The database stores information related to:

- Mines
- Production
- Equipment
- Geological samples

- Weather
 - Satellite observations
 - Blasting
 - Predictions
 - Risk assessments
 - Recommendations
 - Upload history
 - Reserve-potential predictions
-

7.4 Deployment Layer

The prototype has been configured for cloud deployment.

Frontend:

Render

Backend:

Render

Database:

PostgreSQL

Source code:

GitHub

8. Technology Stack

Layer	Technology
Frontend	React
Frontend Build Tool	Vite
Frontend API Client	Axios
Backend	Python
API Framework	FastAPI
Server	Uvicorn
Database	PostgreSQL
Database Driver	psycopg2
Configuration	python-dotenv
Data Processing	Python-based processing
Visualization	Charts + Interactive Map
Version Control	Git
Repository	GitHub
Cloud Hosting	Render

9. Database Design

The PostgreSQL database is named:

sih_26009

The main database entities include:

9.1 mines

Stores basic information about mines.

Typical fields:

- id
 - name
 - location
 - mine_type
 - latitude
 - longitude
 - status
 - created_at
-

9.2 production_records

Stores production information.

Typical information:

- Mine
 - Production date
 - Planned production
 - Actual production
 - Operational delays
 - Downtime
-

9.3 equipment

Stores mine equipment information.

Examples:

- Equipment ID
 - Mine ID
 - Equipment type
 - Status
 - Operational information
-

9.4 equipment_performance

Stores equipment performance indicators.

This data supports equipment risk analysis.

9.5 geological_samples

Stores geological sample information.

Potential information includes:

- Mine
- Sample location
- Geological attributes
- Sample measurements

This information can support reserve-potential analysis.

9.6 weather_data

Stores weather-related observations.

Possible variables include:

- Rainfall
 - Temperature
 - Weather conditions
 - Soil/environmental indicators
-

9.7 satellite_observations

Stores satellite-derived observations.

Possible variables include:

- Vegetation index
 - Soil moisture
 - Land surface temperature
 - Geographic coordinates
 - Observation date
-

9.8 blasting_records

Stores blasting-related operational information.

Possible information:

- Blasting date
- Delay
- Mine

- Operational impact
-

9.9 predictions

Stores model-generated prediction results.

Possible information:

- Prediction date
 - Prediction type
 - Predicted value
 - Model version
 - Confidence information
-

9.10 reserve_potential_predictions

Stores reserve-potential analysis results.

These results can be used to visualize relatively higher potential zones on the map.

9.11 risk_assessments

Stores operational risk information.

Possible information:

- Risk score
 - Risk level
 - Contributing factor
 - Assessment date
-

9.12 recommendations

Stores decision-support recommendations.

Possible information:

- Recommendation
 - Reason
 - Priority
 - Related risk
 - Mine
-

9.13 upload_history

Stores information about uploaded datasets.

Possible information:

- File name
 - Upload time
 - Mine
 - Processing status
 - Inserted rows
-

10. Data Flow

Step 1: Data Collection

Data is collected from:

- Production systems
 - Equipment records
 - Geological information
 - Weather information
 - Satellite observations
 - Blasting records
-

Step 2: Data Ingestion

Users can provide data using:

- File upload
 - Manual entry
-

Step 3: Data Validation

The backend validates incoming information.

Invalid records can be rejected instead of being inserted into the database.

Step 4: Data Storage

Validated information is stored in PostgreSQL.

Step 5: Data Analysis

The backend processes stored information.

Step 6: Prediction and Risk Analysis

The system generates:

- Production estimates
 - Shortfall information
 - Risk scores
 - Equipment risk
 - Reserve-potential results
-

Step 7: Recommendation

The system generates decision-support recommendations based on detected conditions.

Step 8: Visualization

The frontend displays results through:

- KPI cards
 - Charts
 - Risk indicators
 - Equipment status
 - Map
 - Recommendation panel
 - What-if simulation
-

11. Frontend Modules

The frontend dashboard contains the following major modules.

11.1 Top Navigation

Displays:

- Manganese Intelligence
 - AI-Powered Mining Decision Support System
 - Mine information
 - System status
-

11.2 Operational Overview

Displays major project information and operational summary.

11.3 KPI Dashboard

Important KPIs include:

Predicted Production

Shows estimated production.

Latest Shortfall

Shows the latest production gap.

Operational Risk

Shows:

- Risk level
- Risk score

Reserve Potential

Shows the number of identified high-potential zones.

12. Data Ingestion Center

The Data Ingestion Center allows users to add new operational data.

Upload Production File

Supported formats:

- CSV
- XLSX
- XLS
- PDF

The frontend sends the selected file to the backend.

Manual Production Entry

Users can manually enter:

- Production date
 - Planned production
 - Actual production
 - Downtime hours
 - Blasting delay hours
-

Recent Data Activity

The system displays recently processed uploads.

This provides visibility into data ingestion activity.

13. Production Analytics

The Production Overview section displays production information graphically.

The user can compare:

- Planned production
- Actual production
- Production trend

The purpose is to quickly identify production variation.

14. Risk Analysis

Risk Analysis provides a summarized view of operational risk.

Displayed information may include:

- Risk score
- Risk category
- Current risk
- Primary cause

Example:

Risk Score: 32/100
Risk Level: MEDIUM
Primary Cause: High Downtime

These values are prototype outputs and depend on the available dataset.

15. Equipment Risk Module

The equipment module displays equipment-specific risk.

Example:

EQ-001 LOW
EQ-002 HIGH

The purpose is to help identify equipment that may require additional attention.

16. Reserve Potential Map

The system provides an interactive geographical visualization.

The map can display:

- Mine location
- Potential zones
- Geographic coordinates
- Reserve-potential indicators

The map is intended to help users understand the spatial distribution of potential zones.

It does not represent certified geological reserves.

17. AI Recommendations

The recommendation module presents decision-support suggestions.

Examples:

High equipment downtime detected.
Prioritize maintenance for affected equipment.

Blasting delay is contributing to production risk.
Review blasting schedule and operational planning.

Production is below planned level.
Review equipment availability and operational constraints.

Recommendations should be validated by domain experts before operational implementation.

18. What-if Simulation

The What-if Simulation module allows users to test hypothetical operational changes.

Example:

Downtime Reduction
↓
30%
↓
Estimated New Downtime
↓
Estimated Production
↓
Remaining Shortfall

The simulation helps users understand the relationship between operational changes and estimated production outcomes.

It is not a guaranteed forecasting mechanism.

19. Backend API

The backend exposes REST APIs.

Base URL during local development:

`http://127.0.0.1:8000`

Production backend:

`https://sih-26009-backend.onrender.com`

19.1 Health API

GET /api/health

Purpose:

Checks whether the backend service is running.

Example response:

```
{
  "status": "ok",
  "service": "SIH 26009 Backend"
}
```

19.2 Dashboard API

GET /api/dashboard/summary

Provides summarized dashboard information.

Possible outputs:

- Predicted production
 - Risk level
 - Risk score
 - Primary cause
 - High-potential zones
-

19.3 Production API

GET /api/production/

Returns production records.

19.4 Equipment Risk API

GET /api/equipment/risk

Returns equipment risk information.

19.5 Reserve Potential API

GET /api/reserve/potential

Returns reserve-potential records.

19.6 Recommendation API

GET /api/recommendations/

Returns decision-support recommendations.

19.7 What-if Simulation API

GET /api/simulation/what-if?downtime_reduction=30

The parameter represents the hypothetical downtime reduction percentage.

19.8 Upload API

POST /api/ingestion/upload

Used to upload production datasets.

19.9 Manual Entry API

POST /api/ingestion/manual

Used to submit production information manually.

19.10 Upload History API

GET /api/ingestion/history?mine_id=1&limit=20

Returns recent ingestion history.

20. AI/ML Component

The AI/ML layer is intended to support two major intelligence areas.

20.1 Production Prediction

Input variables may include:

- Historical production
- Planned production
- Equipment downtime
- Equipment performance
- Blasting delay
- Weather conditions

Output:

Estimated production

20.2 Reserve Potential Analysis

Potential inputs include:

- Geological sample information
- Geographic location
- Satellite-derived indicators
- Environmental indicators

Output:

Reserve-potential score / zone

21. Risk Assessment Logic

Operational risk can be influenced by multiple factors.

Conceptually:

Equipment Downtime
+
Blasting Delay
+
Weather Impact
+
Production Deviation
↓
Operational Risk

The resulting risk can be categorized into:

LOW
MEDIUM
HIGH

The exact calculation depends on the implemented analytical logic and available data.

22. Security

The application should follow basic software security practices.

Important practices include:

- Store database credentials in environment variables.
 - Do not commit .env files to GitHub.
 - Do not expose database passwords in source code.
 - Validate uploaded files.
 - Validate user input.
 - Restrict database access.
 - Use HTTPS for production APIs.
 - Apply appropriate CORS configuration.
 - Use separate development and production credentials.
 - Rotate credentials if they are accidentally exposed.
-

23. Environment Configuration

Example local environment configuration:

```
DB_HOST=localhost  
DB_PORT=5432  
DB_NAME=sih_26009  
DB_USER=postgres  
DB_PASSWORD=YOUR_POSTGRES_PASSWORD
```

Actual passwords must never be included in source code or public documentation.

24. Local Installation

Step 1: Clone Repository

```
git clone https://github.com/vishal01-code/SIH-26009.git
```

Step 2: Backend

```
cd SIH-26009\backend
```

Create virtual environment:

```
python -m venv .venv
```

Activate:

```
.\.venv\Scripts\Activate.ps1
```

Install dependencies:

```
pip install -r requirements.txt
```

Step 3: Configure Environment

Create:

```
.env
```

Add the required database configuration.

Step 4: Start Backend

```
python -m uvicorn app.main:app --reload
```

Backend:

```
http://127.0.0.1:8000
```

Step 5: Start Frontend

Open another terminal:

```
cd SIH-26009/frontend
```

Install dependencies:

```
npm install
```

Start development server:

```
npm run dev
```

Frontend normally runs on:

```
http://localhost:5173
```

25. Production Deployment

The prototype is configured for cloud deployment.

Frontend

Hosted on Render.

Backend

Hosted on Render.

Database

PostgreSQL.

Source Code

GitHub.

Production frontend communicates with the deployed backend through the configured API base URL.

26. Git Workflow

The project uses Git for version control.

Basic workflow:

`git status`

Add changes:

`git add .`

Commit:

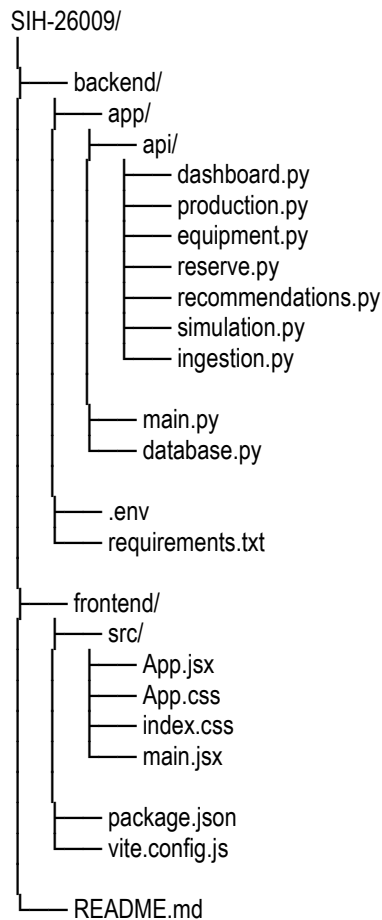
`git commit -m "Update project"`

Push:

`git push origin master`

27. Project Directory Structure

A simplified structure is:



28. Error Handling

The system should handle common errors such as:

Backend unavailable

Display an appropriate API connection message.

Invalid upload

Reject unsupported or invalid files.

Invalid data

Display validation errors.

Empty dataset

Display an appropriate empty-state message instead of breaking the dashboard.

Database error

Log the error on the backend and return an appropriate API response.

Simulation failure

Display an appropriate simulation error and prevent invalid output from being shown as a valid result.

29. Testing Strategy

The project should be tested at multiple levels.

29.1 Unit Testing

Test individual functions such as:

- Shortfall calculation
 - Risk calculation
 - Data validation
 - Recommendation generation
 - Simulation calculation
-

29.2 API Testing

Test:

- Health endpoint
 - Dashboard endpoint
 - Production endpoint
 - Equipment endpoint
 - Reserve endpoint
 - Recommendation endpoint
 - Simulation endpoint
 - Upload endpoint
 - Manual-entry endpoint
-

29.3 Database Testing

Verify:

- Tables exist
 - Foreign keys work
 - Data insertion works
 - Data retrieval works
 - Invalid data is rejected
-

29.4 Frontend Testing

Verify:

- Dashboard loads
 - KPI cards display correct values
 - Charts receive data
 - Map loads
 - File upload works
 - Manual entry works
 - Simulation works
 - Recommendations load
-

29.5 Integration Testing

Verify the complete pipeline:

Frontend
↓
FastAPI
↓
PostgreSQL
↓
Analysis
↓
API Response
↓
Frontend Visualization

30. Sample Operational Scenario

Suppose the mine has:

Planned Production: 500 T
Actual Production: 420 T
Downtime: 8 hours
Blasting Delay: 2 hours

The system may identify:

Production Shortfall: 80 T
Risk Level: High
Primary Cause: High Downtime

The recommendation module may suggest:

Prioritize equipment maintenance and investigate
the source of excessive downtime.

The user can then use the what-if simulation to explore a hypothetical reduction in downtime.

This example is illustrative and does not represent real mine operational data.

31. Advantages

The proposed platform provides:

1. Centralized mining intelligence.
 2. Multi-source data integration.
 3. Production monitoring.
 4. Production shortfall analysis.
 5. Operational risk visibility.
 6. Equipment risk monitoring.
 7. Reserve-potential visualization.
 8. AI/ML-assisted decision support.
 9. Interactive what-if analysis.
 10. Map-based spatial visualization.
 11. Web-based accessibility.
 12. Extensible architecture.
-

32. Limitations

The current prototype has important limitations.

32.1 Synthetic Data

The demonstration currently relies on synthetic/sample data.

32.2 No Real Mine Validation

The model has not been validated against real operational mine data.

32.3 Reserve Potential Is Not Certified Reserves

The reserve module demonstrates potential-zone analysis and does not replace geological exploration or reserve certification.

32.4 Prototype Recommendations

Recommendations are decision-support outputs and require expert verification.

32.5 Model Accuracy

Production and reserve-potential performance depends heavily on the quality, quantity and relevance of training and operational data.

33. Future Scope

Future development can include:

33.1 Real Mining Data Integration

Integrate validated:

- Geological data
- Production data
- Equipment data

- Weather data
 - Satellite data
-

33.2 Advanced Machine Learning

Future versions can evaluate models such as:

- Random Forest
- XGBoost
- Gradient Boosting
- Time-series models
- Neural Networks

The model should be selected based on actual dataset characteristics and validation results.

33.3 Advanced Satellite Analysis

Future versions may incorporate:

- NDVI
 - Soil moisture
 - Land surface temperature
 - Rainfall
 - Remote sensing imagery
 - Multi-temporal satellite observations
-

33.4 GIS Enhancement

Future versions can provide:

- Layer management
 - Geological overlays
 - Satellite layers
 - Mine boundaries
 - Potential-zone heatmaps
 - Spatial filtering
-

33.5 Real-time Monitoring

Integration with mining equipment and IoT systems could allow:

- Real-time equipment status
 - Real-time downtime
 - Sensor monitoring
 - Predictive maintenance
-

33.6 Advanced Predictive Maintenance

Equipment failure prediction can be developed using historical equipment sensor and maintenance data.

33.7 Advanced Recommendation Engine

A future recommendation engine could combine:

- Production forecasts
- Equipment availability
- Weather forecasts
- Blasting schedules
- Maintenance schedules
- Mine plans

to generate more context-aware recommendations.

34. Expected Impact

The proposed system aims to support:

Better Data Visibility

Different mining datasets can be viewed from one platform.

Early Risk Awareness

Potential production risks can be identified before they become larger operational problems.

Better Operational Planning

Production forecasts and what-if analysis can support planning discussions.

Spatial Intelligence

Potential zones can be visualized geographically.

Faster Analysis

A dashboard can reduce the need to manually inspect multiple datasets separately.

Data-driven Decision Support

Mining professionals can use analytical outputs as an additional input to operational decision-making.

35. Target Users

Potential users include:

- Mining engineers
 - Mine managers
 - Production planners
 - Geologists
 - Equipment/maintenance teams
 - Data analysts
 - Mining operations teams
 - Decision-makers
-

36. Non-Functional Requirements

Performance

The dashboard should respond quickly under normal expected workloads.

Reliability

The system should handle invalid input without crashing.

Scalability

The architecture should allow additional mines and datasets to be added in the future.

Security

Credentials and sensitive operational information should be protected.

Maintainability

Backend APIs and frontend modules should remain modular.

Usability

Important operational information should be understandable without requiring users to inspect raw datasets.

37. Functional Requirements

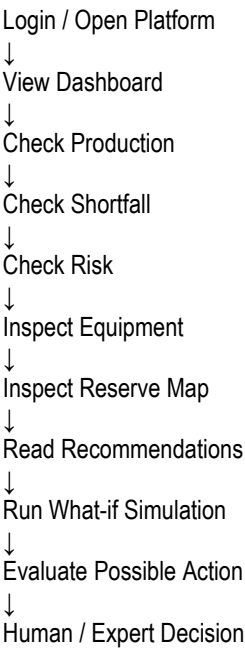
The system should allow users to:

1. View dashboard summary.
2. View production information.
3. Upload production data.
4. Enter production data manually.

- 5. View upload history.
- 6. View production trends.
- 7. View operational risk.
- 8. View equipment risk.
- 9. View reserve-potential zones.
- 10. View recommendations.
- 11. Run what-if simulations.
- 12. Retrieve analytical results through APIs.

38. User Journey

A typical user journey is:



39. Example Dashboard

The dashboard is designed around the following structure:

MANGANESE INTELLIGENCE
AI-POWERED MINING DECISION SUPPORT

Predicted Production	Latest Shortfall	Operational Risk	Reserve Potential
XXX T	XX T	XX/100	X

DATA INGESTION CENTER

Upload Production File Manual Production Entry

Recent Data Activity

PRODUCTION OVERVIEW

Production Chart

RISK ANALYSIS

Risk Score Risk Level Primary Cause

EQUIPMENT RISK

EQ-001 LOW
EQ-002 HIGH

RESERVE POTENTIAL MAP

Interactive Map

AI RECOMMENDATIONS

Recommendation 1
Recommendation 2
Recommendation 3

WHAT-IF SIMULATION

Downtime Reduction: XX%

Estimated Production
Remaining Shortfall

FOOTER

40. Demo Data Policy

The current demonstration uses synthetic/sample data.

Therefore:

- Demonstration values are not real MOIL operational values.
 - Demonstration predictions should not be interpreted as actual mine forecasts.
 - Reserve-potential zones are demonstration outputs.
 - Risk scores are prototype outputs.
 - What-if results are hypothetical estimates.
 - Real-world deployment requires validated datasets, domain expertise, testing and appropriate authorization.
-

41. Project Status

The prototype currently includes the major application structure:

- React frontend
- FastAPI backend
- PostgreSQL database
- Dashboard
- Production module
- Equipment risk module
- Reserve potential module
- Recommendation module
- What-if simulation
- Data ingestion
- Manual production entry
- File upload
- Upload history
- Cloud deployment configuration

The prototype is intended to demonstrate the complete end-to-end concept.

42. Project Repository

GitHub repository:

<https://github.com/vishal01-code/SIH-26009.git>

43. Deployment

Frontend:

<https://sih-26009-frontend.onrender.com>

Backend:

<https://sih-26009-backend.onrender.com>

Health endpoint:

<https://sih-26009-backend.onrender.com/api/health>

44. Team

Team Name

Mine Matrix

Members

1. Vishal Prajapati (Team_Leader)
2. Sachin Gond (Co-leader)
3. Anand Singh
4. Ritu Tiwari
5. Abhishek
6. Saurabh

Project

Manganese Intelligence

SIH Problem Statement

26009

45. Conclusion

Manganese Intelligence demonstrates how AI/ML, mining data analytics and satellite-derived information can be combined into a unified mining decision-support platform.

The system brings together production monitoring, production shortfall analysis, operational risk assessment, equipment risk monitoring, reserve-potential visualization, recommendations and what-if simulation.

The key concept is to transform fragmented mining information into a structured intelligence workflow:

Collect → Process → Analyze → Predict → Assess Risk → Recommend → Simulate → Support Decisions

The current implementation is a prototype using synthetic/sample data. Future development can integrate validated real-world mining and satellite datasets, improve machine-learning models, introduce advanced geospatial analysis and support real-time mining operations.

The final operational decision should always remain with qualified mining professionals and appropriate domain experts.

46. One-Line Project Definition

Manganese Intelligence is an AI/ML-powered mining decision-support platform that combines geological, production, equipment, weather and satellite-derived data to analyze reserve potential, predict production shortfalls, assess operational risks and support data-driven mining decisions.